

Oracle HR Schema – Subqueries Analytical Reports

Objective

Develop analytical reports using subqueries in the Oracle HR Schema while demonstrating an understanding of business requirements, query execution, and different subquery techniques.

Salary Analysis Using Subqueries

1. Employees Earning More Than the Company Average Salary

Business Requirement

Identify employees whose salaries are higher than the overall company average salary.

```
SELECT employee_id,  
       first_name,  
       last_name,  
       salary  
FROM employees  
WHERE salary >  
       (SELECT AVG(salary)  
        FROM employees);
```

Subquery Type

Scalar Subquery

Execution Logic

1. The subquery calculates the average salary of all employees.
 2. A single value is returned.
 3. The outer query compares each employee's salary against that value.
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2. Employees Earning Less Than the Company's Highest Salary

Business Requirement

Find employees whose salaries are lower than the highest salary in the company.

```
SELECT employee_id,  
       first_name,  
       last_name,  
       salary  
FROM employees  
WHERE salary <  
       (SELECT MAX(salary)  
        FROM employees);
```

Subquery Type

Scalar Subquery

Execution Logic

1. The subquery finds the highest salary in the company.
 2. The outer query returns employees whose salaries are below that amount.
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3. Departments Whose Average Salary Exceeds the Company Average Salary

Business Requirement

Identify departments where the average salary is higher than the company's overall average salary.

```
SELECT department_id
FROM employees
GROUP BY department_id
HAVING AVG(salary) >
      (SELECT AVG(salary)
       FROM employees);
```

Subquery Type

Scalar Subquery

Execution Logic

1. Calculate the company-wide average salary.
 2. Calculate the average salary for each department.
 3. Return departments whose average salary exceeds the company average.
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4. Employees Earning the Highest Salary Within Their Department

Business Requirement

Identify employees who earn the highest salary in their department.

```
SELECT employee_id,  
       first_name,  
       department_id,  
       salary  
FROM employees e  
WHERE salary =  
      (  
        SELECT MAX(salary)  
        FROM employees  
        WHERE department_id = e.department_id  
      );
```

Subquery Type

Correlated Subquery

Execution Logic

1. For each employee, Oracle identifies the employee's department.
 2. The subquery finds the maximum salary within that department.
 3. Employees matching that maximum salary are returned.
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Department & Location Analysis

1. Employees Working in Departments Located in the United States

Business Requirement

Find employees working in departments located in the United States.

```
SELECT employee_id,  
       first_name,  
       last_name  
FROM employees  
WHERE department_id IN  
(  
  SELECT department_id  
  FROM departments  
  WHERE location_id IN  
    (  
      SELECT location_id  
      FROM locations  
      WHERE country_id = 'US'  
    )  
);
```

Subquery Type

Nested Subquery

Execution Logic

US → Locations → Departments → Employees

2. Departments That Currently Contain Employees

Business Requirement

List departments that currently have employees assigned to them.

```
SELECT department_id,  
       department_name  
FROM departments  
WHERE department_id IN  
(  
    SELECT department_id  
    FROM employees  
);
```

Subquery Type

Multi-Row Subquery

Execution Logic

The subquery returns multiple department IDs from the Employees table, and the outer query retrieves matching departments.

3. Departments That Currently Have No Employees

Business Requirement

Identify departments without any employees.

```
SELECT department_id,  
       department_name  
FROM departments  
WHERE department_id NOT IN  
(  
    SELECT department_id  
    FROM employees  
    WHERE department_id IS NOT NULL  
);
```

Subquery Type

Multi-Row Subquery

Execution Logic

The subquery returns departments containing employees. The outer query excludes them.

4. Employees Working in Departments Located in the Europe Region

Business Requirement

Find employees working in departments located in the Europe region.

```
SELECT employee_id,  
       first_name,  
       last_name  
FROM employees  
WHERE department_id IN  
(  
    SELECT department_id  
    FROM departments  
    WHERE location_id IN  
    (  
        SELECT location_id  
        FROM locations  
        WHERE country_id IN  
        (  
            SELECT country_id  
            FROM countries  
            WHERE region_id =  
            (  
                SELECT region_id  
                FROM regions  
                WHERE region_name = 'Europe'  
            )  
        )  
    )  
);
```

Subquery Type

Nested Subquery

Execution Logic

Europe Region → Countries → Locations → Departments → Employees

Manager & Hierarchy Analysis

1. Employees Earning More Than Their Manager

Business Requirement

Find employees whose salary is higher than their manager's salary.

```
SELECT employee_id,  
       first_name,  
       salary  
FROM employees e  
WHERE salary >  
(  
    SELECT salary  
    FROM employees  
    WHERE employee_id = e.manager_id  
);
```

Subquery Type

Correlated Subquery

Execution Logic

For each employee, Oracle retrieves the manager's salary and compares it with the employee's salary.

2. Managers Supervising at Least One Employee Earning More Than 10,000

Business Requirement

Identify managers who supervise employees earning more than 10,000.

```
SELECT employee_id,  
       first_name  
FROM employees  
WHERE employee_id IN  
(  
    SELECT manager_id  
    FROM employees  
    WHERE salary > 10000  
);
```

Subquery Type

Multi-Row Subquery

3. Employees Whose Manager Works in the Same Department

Business Requirement

Find employees whose manager belongs to the same department.

```
SELECT employee_id,  
       first_name  
FROM employees e  
WHERE department_id =  
(  
    SELECT department_id  
    FROM employees  
    WHERE employee_id = e.manager_id  
);
```

Subquery Type

Correlated Subquery

4. Employees Whose Manager Works in a Different Department

Business Requirement

Find employees whose manager belongs to a different department.

```
SELECT employee_id,  
       first_name  
FROM employees e  
WHERE department_id <>  
      (  
        SELECT department_id  
        FROM employees  
        WHERE employee_id = e.manager_id  
      );
```

Subquery Type

Correlated Subquery

Inline View & Query Execution

1. Employees Earning More Than 10,000

Business Requirement

Display employees whose salaries exceed 10,000 using an inline view.

```
SELECT *
FROM
(
    SELECT employee_id,
           first_name,
           salary
    FROM employees
) emp_view
WHERE salary > 10000;
```

Subquery Type

Inline View

Execution Logic

The inline view creates a temporary result set, which is then filtered by the outer query.

2. Departments with Average Salary Greater Than 9,000

```
SELECT *
FROM
(
    SELECT department_id,
           AVG(salary) avg_salary
    FROM employees
    GROUP BY department_id
) dept_avg
WHERE avg_salary > 9000;
```

Subquery Type

Inline View

3. Departments with More Than Five Employees

```
SELECT *
FROM
(
    SELECT department_id,
           COUNT(*) emp_count
    FROM employees
    GROUP BY department_id
) dept_count
WHERE emp_count > 5;
```

Subquery Type

Inline View

Advanced Subquery Logic

1. Employees Earning Above Their Department Average Salary

Business Requirement

Identify employees earning more than the average salary of their department.

```
SELECT employee_id,
       first_name,
       salary
FROM employees e
WHERE salary >
(
    SELECT AVG(salary)
    FROM employees
    WHERE department_id = e.department_id
);
```

Subquery Type

Correlated Subquery

2. Employees Earning Above the Average Salary for Their Job Role

Business Requirement

Find employees whose salaries are above the average salary of employees with the same job role.

```
SELECT employee_id,  
       first_name,  
       salary,  
       job_id  
FROM employees e  
WHERE salary >  
(  
    SELECT AVG(salary)  
    FROM employees  
    WHERE job_id = e.job_id  
);
```

Subquery Type

Correlated Subquery

3. Departments Where All Employees Earn More Than 5,000

Business Requirement

Find departments where every employee earns more than 5,000.

```
SELECT department_id  
FROM employees  
GROUP BY department_id  
HAVING MIN(salary) > 5000;
```

Subquery Type

Aggregate Subquery Logic

Execution Logic

If the minimum salary in a department is greater than 5,000, then every employee in that department earns more than 5,000.

4. Employees Who Are the Only Employee in Their Department

Business Requirement

Identify employees who are the sole employee in their department.

```
SELECT employee_id,  
       first_name,  
       department_id  
FROM employees e  
WHERE 1 =  
(  
    SELECT COUNT(*)  
    FROM employees  
    WHERE department_id = e.department_id  
);
```

Subquery Type

Correlated Subquery

Executive HR Analytics Challenge

1. Employees Earning Above the Company Average Salary

```
SELECT *  
FROM employees  
WHERE salary >  
(  
    SELECT AVG(salary)  
    FROM employees  
);
```

Subquery Type

Scalar Subquery

2. Departments with More Employees Than Department 50

```
SELECT department_id
FROM employees
GROUP BY department_id
HAVING COUNT(*) >
(
    SELECT COUNT(*)
    FROM employees
    WHERE department_id = 50
);
```

Subquery Type

Scalar Subquery

3. Employees Working in the Department with the Highest Average Salary

```
SELECT *
FROM employees
WHERE department_id =
(
    SELECT department_id
    FROM
    (
        SELECT department_id,
               AVG(salary) avg_sal
        FROM employees
        GROUP BY department_id
        ORDER BY avg_sal DESC
    )
    WHERE ROWNUM = 1
);
```

Subquery Type

Inline View + Nested Subquery

4. Employees Earning the Second-Highest Salary

```
SELECT *  
FROM employees  
WHERE salary =  
(  
    SELECT MAX(salary)  
    FROM employees  
    WHERE salary <  
        (  
            SELECT MAX(salary)  
            FROM employees  
        )  
);
```

Subquery Type

Nested Scalar Subquery

Execution Logic

1. Find the highest salary.
 2. Exclude that salary.
 3. Find the maximum of the remaining salaries.
 4. Return employee(s) earning that amount.
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5. Managers Whose Employees' Average Salary Exceeds 10,000

```
SELECT employee_id,  
       first_name  
FROM employees m  
WHERE employee_id IN  
(  
    SELECT manager_id  
    FROM employees  
    GROUP BY manager_id  
    HAVING AVG(salary) > 10000  
);
```

Subquery Type

Multi-Row Subquery

Execution Logic

1. Group employees by manager.
 2. Calculate the average salary of each manager's team.
 3. Keep managers whose team's average salary exceeds 10,000.
 4. Return the manager details from the Employees table.
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Summary of Subquery Types Used

<i>Subquery Type</i>	<i>Description</i>
<i>Scalar Subquery</i>	Returns a single value (AVG, MAX, MIN, COUNT)
<i>Multi-Row Subquery</i>	Returns multiple rows, commonly used with IN or NOT IN
<i>Correlated Subquery</i>	Depends on values from the outer query
<i>Nested Subquery</i>	A subquery inside another subquery
<i>Inline View</i>	A subquery placed in the FROM clause
<i>Aggregate Subquery</i>	Uses aggregate functions such as AVG, MAX, MIN, and COUNT